

Coordinator



Observes Public state s_t
and agent's action $a_t^1, \dots, a_t^n$



Does NOT observe agents'
hidden personas θ_i



Learn and plans to
maximize team reward

State s_t
(Public)

Action $a_t^1, \dots, a_t^n$
(observed)

Reward R_t
(observed)



Multi-Agent Systems

persona θ_1



Altruistic Builder

Cooperative and
pro-social;

Example: "I'll help
build it.."

Action a_t^1

persona θ_2



Free Rider

Prefers to benefit
without contributing
effort;

Example: "Let other
do this work.."

Action a_t^2

persona θ_n



Strategic Hedger

Waits to observe
others and balance
risk and reward;

Example: "I'll act
after I see others.."

Action a_t^n

Joint Transition $T(s_{t+1} | s_t, a_t^1, \dots, a_t^n)$ & Joint Reward $R(s_t, a_t^1, \dots, a_t^n; \theta_1, \dots, \theta_n)$

PACT Pipeline: Inference Outside the Prompt, Planning inside the Loop

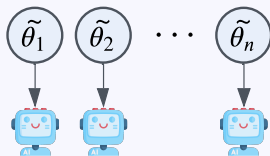
1. Persona Set & Priors

For each agent i , maintain a
candidate set Θ_i and a prior $P_i(\theta_i)$

Agent i	Prior P_i
θ_1	
θ_2	
$\vdots$	$\vdots$
$\theta_{ \Theta_i }$	

2. Sample Joint Personas

Sample one persona $\tilde{\theta}_i \sim P_i(\theta_i)$
for each agent independently



This sampled profile defines the
model for planning

3. Plan & Execute

For each agent i , maintain a
candidate set Θ_i and a prior $P_i(\theta_i)$



Planner

$\pi^*(\tilde{\theta}_1, \dots, \tilde{\theta}_n)$

Pact+

$Q_h^+(s, a) = Q_h(s, a) + \beta D_k(s, a)$

4. Score Observed Actions

Compute likelihood of observed action
under every candidate's personas.

Observed
action
 $a_{k,h}^i$

Prompt $(\theta_1^i) \rightarrow \ell_1^i$
Prompt $(\theta_2^i) \rightarrow \ell_2^i$
 $\vdots$
Prompt $(\theta_t^i) \rightarrow \ell_t^i$

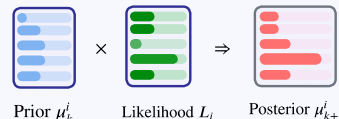
$\ell_{i,h}(\theta_i) = P_{LLM}(\phi(a_{i,h}) | c_h, \text{prompt}(\theta_i))$

$L_i(\tau_k; \theta_i^j) = \prod_{h=1}^H \ell_{i,h}(\theta_i^j)$

5. Bayesian Posterior Update

Update posterior over each agent's
personas using likelihoods

$$\mu_{k+1}^i(\theta_i^j) = \frac{\mu_k^i(\theta_i^j) L_i}{\sum_{\theta_i^q \in \Theta_i} \mu_k^i(\theta_i^q) L_i}$$



$\mu_{k+1}^\Theta(\theta) = \prod_i \mu_{k+1}^i(\theta_i)$